

P-2104B Return-to-Service Assessment

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Recommendation

Pump P-2104B must not be returned to service on its existing seal arrangement. Vibration has risen from 3.9 mm/s to 9.6 mm/s over eight weeks, crossing the 7.1 mm/s alert threshold on 2026-08-13 [VIB-CDU2-2026_Vibration_Log.xlsx, sheet Readings].

Evidence

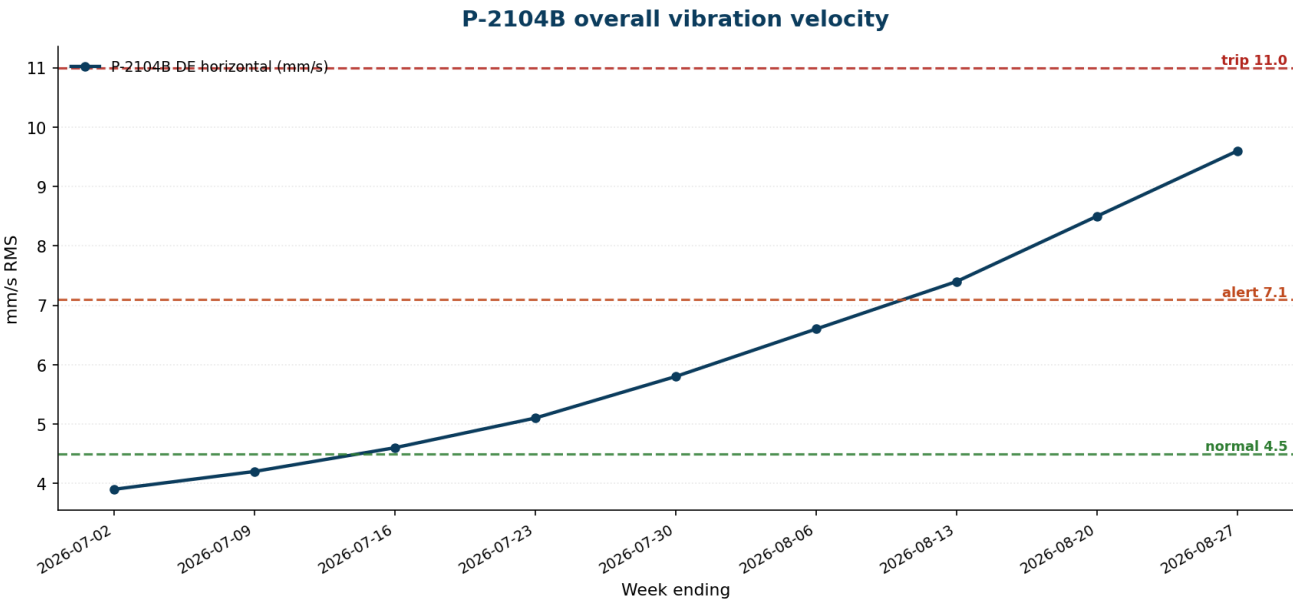
- Overall velocity reached 9.6 mm/s on 2026-08-27, which is 87% of the 11.0 mm/s trip limit [VIB-CDU2-2026_Vibration_Log.xlsx, sheet Limits].
- The August seal failure on the same machine was attributed to Plan 52 pot level loss [INC-2026-0042_Seal_Failure_P-2104B.pdf, p.2].
- HAZOP action H7-03 covering seal-pot low-level alarms remains OVERDUE [HAZOP-CDU2-2024_Node7_Action_Register.pdf, p.1].

Blockers

- Seal kit SK-682-2104-A3 shows zero on hand with a 45-day lead time [SPR-ROT-001_Spares_Inventory.xlsx, sheet Stock].
- Hot work in CDU-2 is suspended until 2026-11-30 under circular SC-2026-11 [SC-2026-11_Hot_Work_Restriction_CDU2.pdf, p.1].
- Two operators have lapsed competency for the changeover SOP [TRN-CDU2-MATRIX_Competency_Matrix.xlsx, sheet Matrix].

Next steps

Raise an expedite request against the seal kit, close H7-03 before restart, and run the changeover on P-2104A per SOP-CDU2-014 Rev 4 step 7 [SOP-CDU2-014_Rev4_Pump_Changeover.pdf, p.3].



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PROVENANCE AND CHAIN OF CUSTODY

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Model	qwen2.5:7b-instruct served locally at http://127.0.0.1:11434
Question	Can we return P-2104B to service, and what is blocking the seal upgrade?
Sources cited	6 document reference(s)
Audit chain	33 records, intact, head 715905f009c81ca900d7d97080e0ad8c
Session	4194ed5e9f57

Sources

- 1. VIB-CDU2-2026_Vibration_Log.xlsx, sheet Readings
- 2. INC-2026-0042_Seal_Failure_P-2104B.pdf, p.2
- 3. HAZOP-CDU2-2024_Node7_Action_Register.pdf, p.1
- 4. SPR-ROT-001_Spares_Inventory.xlsx, sheet Stock
- 5. SC-2026-11_Hot_Work_Restriction_CDU2.pdf, p.1
- 6. SOP-CDU2-014_Rev4_Pump_Changeover.pdf, p.3